

Econ 139: O*NET

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O*NET

- O*NET (Occupational Information Network) is a database run by the Department of Labor
- Overall goal: provide a standardized way to understand what individuals do in different occupations
 - This breaks occupations into work activities, tasks, skills, and work requirements
- Intended to be used by many different types of people: not just researchers, but job seekers, firms, etc.



O*NET Online

- O*NET Online is a simple website where you can search through occupations
 - Allows you to get some clear information. For example, if you are a job seeker or firm, you can see generally what activities are performed by this occupation
 - However, researchers tend to use the raw data files
 - Allows you to study many occupations at once
 - Gives you access to much more information not available on the website
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O*NET Raw Data Files

- The O*NET raw data files are quite complex
 - For the most recent download, it came with around 30 files
 - Depending on what you want to do, you need to link these files together
 - Often the content of files covers similar topics, so it can quickly get confusing
- We are going to focus our discussion on a single file the Work Activities File.

O*NET Raw Data Files

- We will study two types of questions in the Work Activities file
- The first are labelled **IMPORTANCE (IM)** questions
- These questions ask how important a given activity is for a job

***Getting information** includes observing, receiving, and otherwise obtaining information from all relevant sources.*

1. How important is getting information to the performance of your current job?

- ☐ Not important → **Go to 2**
- ☐ Somewhat important
- ☐ Important
- ☐ Very important
- ☐ Extremely important

O*NET Raw Data Files

- We will study two types of questions in the Work Activities file
- The second set are labelled **LEVEL (LV)** questions
- These questions ask for some measure of the level needed to perform some activity

►1b. If at least somewhat important, what level of complexity of getting information is needed to perform your current job?

Examples of activities from a variety of jobs:

Low: Follow a standard blueprint

Moderate: Review a budget

High: Study international tax laws

☐ 1 Low	
☐ 2	
☐ 3	
☐ 4 Moderate	
☐ 5	
☐ 6	
☐ 7 High	

Variables in Work Activities File

- Here is a table of variables in the Work Activities File
- We will discuss a subset of these

O*NET-SOC Code	Element ID	Element Name
Scale ID	Data Value	N
Standard Error	Lower CI Bound	Upper CI Bound
Recommend Suppress	Not Relevant	Date
Domain Source		

Structure of O*NET Work Activities

- The unit-of-observation in the Work Activities file is a bit more complicated than previous data analysis
 - CPS, for example, the unit-of-observation was a person
 - This means every row corresponded to a unique person
- Each row in the dataset tells us information about a specific question that was answered by individuals from a specific occupation

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- Element ID and Element Name code for a specific work activity (i.e. a specific question)
 - For example our “Getting Information” question Element ID is 4.A.1.a.1 and the name is Getting Information
- Scale ID tells us if the question corresponds to the importance question (IM) or the level question (LV)

Analyzing a Single Question

- To drill down, let's focus on a single question (4.A.1.a.1) and a single scale (IM)
- This will retrieve the (average) answer to the Getting Information question for the 894 unique occupations in the data

Analyzing a Single Question

Value of 1 corresponds to not important

Value of 5 corresponds to Extremely Important

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Statistic	Value
Mean	4.22
10th Percentile	3.71
75th Percentile	4.48
<i>Number of Observations</i>	<i>894</i>

Analyzing a Single Question

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Value of 5 corresponds to Extremely Important

So an average of 4.22 implies on average across all occupations, getting information is deemed very important

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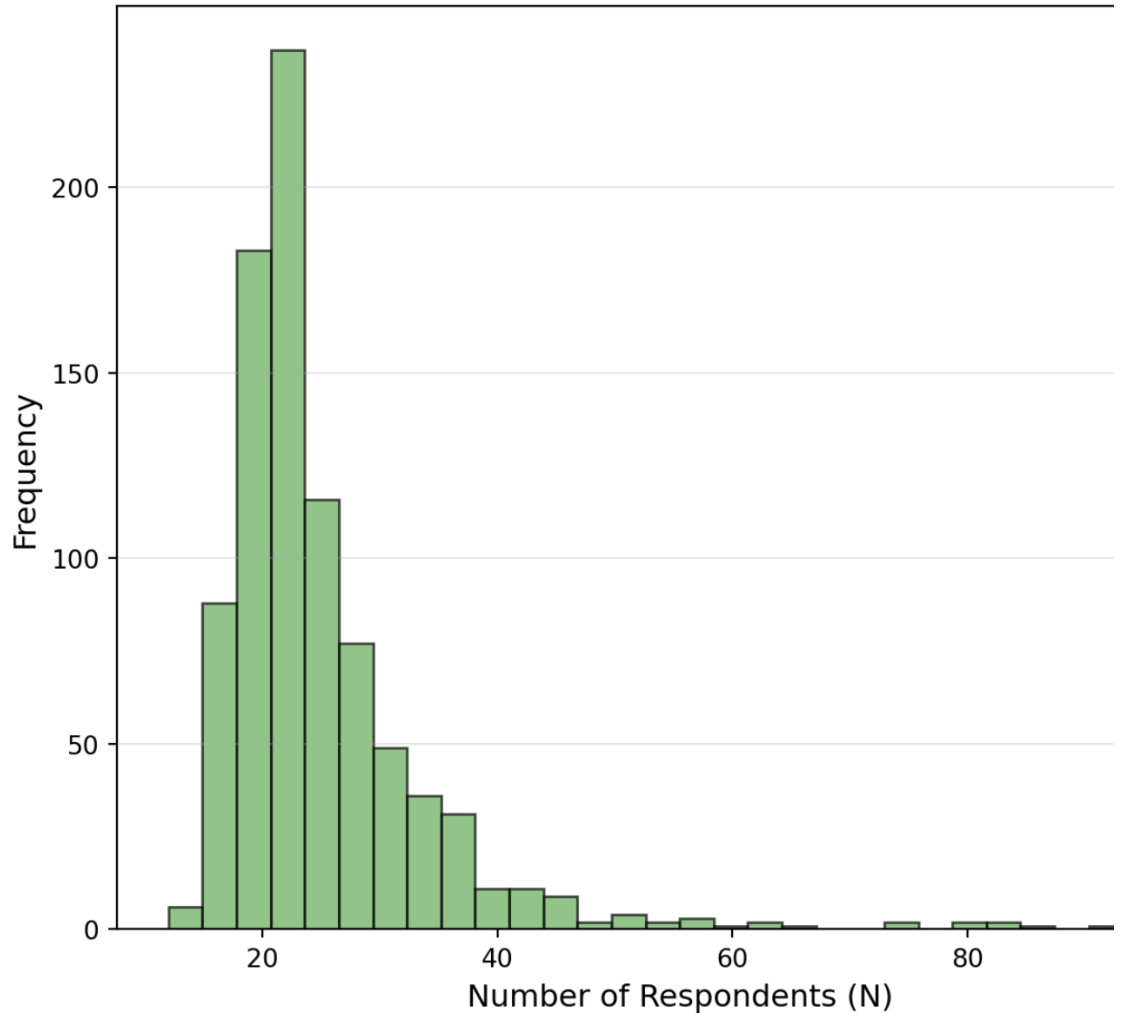
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Number of Respondents per Occupation (N)

- Most occupations do not have an enormous number of respondents
 - This is why O*NET provides standard errors for estimates
- We won't focus on standard errors in this class, but good to keep in mind that sample size per occupation is relatively small

Distribution of Number of Respondents Across Occu



Analyzing a single occupation

- O*NET allows you to again focus on a single occupation
- Let's look at security guards and see what are the most important tasks

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33-9032.00	Identifying Objects, Actions, and Events	4.76	16
33-9032.00	Getting Information	4.58	16
33-9032.00	Communicating with Supervisors, Peers, or Subordinates	4.54	16
33-9032.00	Documenting/Recording Information	4.48	16
33-9032.00	Establishing and Maintaining Interpersonal Relationships	4.45	16

Analyzing a single occupation

- You will see these as the top 5 work activities in O*NET online if you search for security guards
- But getting the raw data allows you to understand the context much better
 - Why are the tasks ordered the way they are
 - How many people is this ordering based on
 - On average, security guards rate these tasks between very important and extremely important

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